



Searching in the sand: Protracted video deficit in U.S. preschoolers' spatial recall using a continuous search space

Heather Kirkorian | Vanessa Simmering

University of Wisconsin-Madison, Madison, Wisconsin, USA

Correspondence

Heather Kirkorian, Department of Human Development and Family Studies, University of Wisconsin-Madison, Madison, WI 53706, USA.
Email: kirkorian@wisc.edu

Abstract

Young children exhibit a video deficit for spatial recall, learning less from on-screen than in-person demonstrations. Some theoretical accounts emphasize memory constraints (e.g., insufficient retrieval cues, competition between memory representations). Such accounts imply memory representations are graded, yet video deficit studies measuring spatial recall operationalize memory retrieval as dichotomous (success or failure). The current study tested a graded-representation account using a spatial recall task with a continuous search space (i.e., sandbox) rather than discrete locations. With this more sensitive task, a protracted video deficit for spatial recall was found in children 4–5 years old ($n = 51$). This may be due to weaker memory representations in the screen condition, evidenced by higher variability and greater perseverative bias. In general, perseverative bias decreased with repeated trials. The discussion considers how the results support a graded-representation account, potentially explaining why children might exhibit a video deficit in some tasks but not others.

KEYWORDS

graded representations, memory retrieval, object retrieval, proactive interference, spatial recall, video deficit

Research Highlights

- The task used a continuous search space (sandbox), making it more difficult and sensitive than spatial recall tasks used in prior video deficit research.
- Spatial recall among 4- and 5-year-old children was more variable after watching hiding events on screen via live video feed than through a window.
- Children's spatial recall from screens was more susceptible to proactive interference, evidenced by more perseverative bias in an A-not-B design.
- The results demonstrate memory representations blend experiences that accumulate over time and explain why the video deficit may be protracted for more difficult tasks.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2023 The Authors. *Developmental Science* published by John Wiley & Sons Ltd.



1 | INTRODUCTION

Transferring information from one context to another is fundamental to learning (Barnett & Ceci, 2002). Yet, transfer poses unique challenges for infants and young children, as when transferring from video to real-life contexts (Hipp et al., 2017). Numerous studies have documented a “video deficit”, or a domain-general tendency for toddlers to learn less from a video demonstration than the same demonstration presented in person (Anderson & Pempek, 2005; Barr, 2010, 2013; Troseth, 2010). A video deficit has been observed for many knowledge domains, including spatial recall (Schmitt & Anderson, 2002; Troseth & DeLoache, 1998), word learning (Krcmar et al., 2007; Roseberry et al., 2009), self-recognition (Suddendorf et al., 2007), and phoneme perception (Kuhl et al., 2003).

Several not-mutually-exclusive accounts have been proposed (see Strouse & Samson, 2021). Some accounts posit that video (compared to in-person) demonstrations produce relatively weak memory representations, which is consistent with the view that representations are graded in nature, rather than merely present or absent (Munakata, 2001). Spatial recall tasks provide particularly strong support for this hypothesis by measuring perseverative bias when children search on multiple back-to-back trials. Yet, these video deficit studies operationalize memory retrieval as dichotomous, such as finding (vs. failing to find) a toy hidden in one of four discrete locations. Such tasks fail to capture gradations in children’s memory representations. Moreover, external memory supports (e.g., nameable landmarks) may simplify tasks in ways that underestimate a video deficit among older children. Thus, we test a graded representations account of the video deficit using a task that can show more continuous interaction between memory for different locations.

1.1 | Current accounts of the video deficit

Some accounts of the video deficit are not specific to video demonstrations, instead explaining generally poor transfer by infants and toddlers in a wide range of contexts. One such account posits that infants have poor representational flexibility, preventing infants from retrieving memories when the perceptual features change from one context to another (Barr, 2013; Hipp et al., 2017). Another such account posits that infants lack symbolic thinking, leading them to hold separate, competing memory representations of on-screen versus in-person experiences (Troseth, 2010). Consistent with these general transfer accounts, the video deficit typically declines by age 36 months (Strouse & Samson, 2021), has a similar developmental time course as transfer using other symbolic artifacts (e.g., photographs, miniature 3D scale models), and is reduced when toddlers have naturalistic experience with symbolic media (Troseth, 2003; Troseth et al., 2007). As further evidence, toddlers exhibit a general “transfer deficit”, performing less well when transferring across dimensions in either direction (2D → 3D or 3D → 2D) than within a single dimension (2D → 2D or 3D → 3D) (Zack et al., 2009).

Other accounts posit that video demonstrations are unique. Such accounts argue that video is perceptually impoverished, providing less information than real-life experience. As a result, video produces weaker memory representations that are less flexible and more susceptible to proactive interference (Barr, 2013; Schmidt et al., 2007). These accounts may explain why toddlers appear to process objects more slowly when presented on-screen versus in-person (Carver et al., 2006; Kirkorian et al., 2016) and why the video deficit is reduced with repeated exposure to video demonstrations (Barr et al., 2007; Krcmar, 2010; Strouse & Troseth, 2008).

A common thread across several accounts of the video deficit is the role of memory constraints, including mismatched retrieval cues and competition between multiple representations when transferring information from one context or medium to another (e.g., 2D to 3D). To the extent that such memory constraints play a role, the video deficit may be sensitive to task demands like memory load. That is, if memory representations are graded in nature, children may succeed at relatively simple tasks but fail at more difficult tasks that measure the same knowledge but require a stronger memory representation (Munakata, 2001). Indeed, although most studies report a video deficit before age 36 months (Strouse & Samson, 2021), some tasks reveal a video deficit among preschoolers (Dickerson et al., 2013; Flynn & Whiten, 2008; Reiß et al., 2019).

1.2 | Spatial recall as a unique test case

The largest video deficit effects have been observed using spatial recall tasks in which children search for a hidden object based on information they observed (e.g., watching the experimenter hide the object) or heard (e.g., “I hid the toy behind the couch.”) (Strouse & Samson, 2021). Such tasks have been used to test children’s transfer from a wide range of stimuli, including picture books, videos, touchscreen applications, and miniature 3D scale models (Troseth, 2010; Troseth et al., 2019). In video deficit studies, toddlers search for an object hidden in discrete, nameable locations (e.g., behind a chair, under a pillow). Thus, individual search trials are scored dichotomously as correct or incorrect. Such studies reveal a video deficit that declines during the third year after birth (Kirkorian et al., 2016; Schmidt et al., 2007; Schmitt & Anderson, 2002; Troseth & DeLoache, 1998). For instance, Schmitt and Anderson reported that toddlers’ object retrieval using video hiding events increased from 23% for 24-month-olds to 56% for 30-month-olds and 82% for 36-month-olds. By comparison, performance was 85% or higher for each age group when toddlers viewed hiding events in person (Schmitt & Anderson, 2002).

Spatial recall tasks pose unique challenges for memory retrieval. Testing children with the same stimuli across multiple trials can introduce proactive interference from previous search trials. That is, what children saw and did on earlier trials can interfere with later trials. For this reason, children often demonstrate higher performance on the first search trial than on subsequent trials (Sharon & DeLoache, 2003). This decline in performance is due to perseveration, or searching the



location that was correct on the previous trial rather than the current one.

Perseveration is more common when children view hiding events on video than in person (Kirkorian et al., 2016; Schmidt et al., 2007; Schmitt & Anderson, 2002; Suddendorf, 2003; Troseth, 2010). Removing the opportunity to perseverate by testing spatial recall in a different room on each trial improved 2-year-olds' spatial recall and reduced the video deficit (Suddendorf, 2003). On the other hand, presenting conflicting demonstrations in person and then on video on the same trial increased 3-year-olds' perseveration following video demonstrations (Zelazo et al., 1999). Together, these findings are consistent with hypotheses that memory representations based on videos compete with, and are weaker than, those based on in-person events (Barr, 2013; Schmidt et al., 2007; Troseth, 2010).

Although the video deficit for spatial recall typically disappears by age 36 months (Schmitt & Anderson, 2002), prior spatial recall studies operationalize memory retrieval as dichotomous, measuring success or failure on a given trial. Such studies mask potential gradations in children's memory representations and, as a result, may underestimate a video deficit among older children. Indeed, studies using other tasks suggest a video deficit may persist well beyond 36 months. For instance, a video deficit for action imitation declined by age 36 months in one study using relatively simple action sequences (McCall et al., 1977) but persisted in older preschoolers using action sequences that involved more steps (Flynn & Whiten, 2008) or distractors (Dickerson et al., 2013). Similarly, a video deficit was observed in 4- and 5-year-old children using a false belief task (Reiß et al., 2019), which most children master by age 4.5 years (Wellman et al., 2001). Such findings suggest task difficulty influences the developmental time course of the video deficit (Barr, 2010; Kirkorian, 2018). Specifically, we posit the relatively simple tasks used in past research require only weak memory representations and mask a video deficit in older children.

1.3 | Measuring graded memory representations

The spatial recall tasks typically used to study the video deficit rely on discrete, nameable hiding locations (e.g., behind the chair, under the pillow). Performance is scored dichotomously on each trial: finding the toy or not. Such tasks may mask variability in children's transfer by providing external cues that can support searching from a weak memory representation and overlooking continuous gradations in memory representations. As noted earlier, it is common in these tasks for children to make perseverative errors, such as searching in the hiding location from the first trial on the second trial. This type of response indicates that the memory representation for the first location was stronger than the second, but it cannot provide further detail on the relative strength of the representations. For example, it could be that the second location was entirely forgotten, and a weak memory from the first search event served to drive the response to the first location. Alternatively, both representations could be relatively strong, with the memory for the first location being only slightly stronger than the first. The same is true for cases when a child responds correctly: The memory represen-

tation of the correct location on the second trial could be only slightly stronger than memory from the first trial, but the discrete nature of the task masks such variation in the strength of memory representations. In sum, the discrete hiding locations and dichotomized scoring of responses could make the difference between incorrect and correct responding appear more distinct than the difference in memory representations actually is.

A task without discrete hiding locations allows continuous measurement of error and could reveal more graded differences in memory representations of on-screen versus in-person events. Such tasks have already been used to reveal subtle age-related improvements in memory. For instance, consider research assessing spatial recall by having children watch an experimenter bury a toy in a long sandbox rather than a few discrete locations. This sandbox task has been used in research with children 2–11 years old (Simmering & Schutte, 2016). The sandbox task presents a uniform search space, requiring children to encode locations relative to the edges and midline symmetry axis of the sandbox rather than encoding visually distinct, easily nameable locations. Moreover, this task provides a more sensitive measure of search error because responses are measured as amount of error on a continuous scale (distance from target) rather than as dichotomous responses (correct/incorrect).

Using the sandbox task, Spencer et al. (2001) designed a variation of the A-not-B task (Piaget, 1954) to test for preservation in toddlers. In their design, participants watched an experimenter bury a toy in the sand, then searched for it after a short delay. The A-not-B design included repeated trials to the first "A" location, followed by trials to a nearby "B" location. Toddlers' searches showed systematic bias toward the A location on B trials, indicating the memory they built up of the A location across repeated trials combined with their memory of the B location to guide their search. Put differently, children's responses on B trials did not reflect a binary choice between two discrete representations (one for A and another for B) but rather a sample drawn from a continuous distribution of possible locations that built up over time, incorporating their past experience with both the A and B locations. Schutte et al. (2003) tested how the influence of memory for nearby locations changed during early childhood and found that 4- and 6-year-old children showed similar biases, but at smaller distances between the A and B locations with age. These results show that the A-not-B design of the sandbox task provides a measure of continuous interactions between memory representations during early childhood.

1.4 | The current study

Given the sensitivity of the sandbox task to continuous variability and perseverative bias, it is well suited to probing for gradation in the video deficit in young children's spatial recall. In the current study, 51 children (4–5 years) completed the sandbox task, observing hiding events either through a window or on screen via live video feed. To the extent video demonstrations result in weaker memory representations, we expected a video deficit effect such that spatial recall from on-screen (vs. in-person) hiding events was more variable and more susceptible to

perseverative bias due to proactive interference from previous search trials (i.e., systematic error on B trials toward the A location). To the extent memory representations are graded in nature, we expected this perseverative bias to decrease gradually, with children's searches moving steadily closer to the B location across trials rather than reflecting a binary choice between the A and B locations.

2 | METHOD

2.1 | Participants

This study was approved by the Institutional Review Board for Education and Social/Behavioral Science at the University of Wisconsin-Madison. Parents provided written informed consent for their child to participate, and children provided verbal assent. Participants were recruited through a database compiled by a university-affiliated research center in Madison, WI, a small US city. We recruited a convenience sample of 56 children ($M_{\text{age}} = 4.82$ years, $SD_{\text{age}} = 0.50$ years, $\text{range}_{\text{age}} = 4.13\text{--}5.89$ years; 19 girls, 37 boys). Most parents (84%) identified their child's race and ethnicity as white and non-Hispanic; the remaining were white and Hispanic (5%), Asian or Asian American (5%), Black or African American (2%), or another race (4%). Parents also reported maternal education, with the majority (88%) reporting at least a 4-year college degree. Five participants were excluded from analyses due to noncompliance ($n = 2$) or incomplete data ($n = 3$). The remaining 51 children were assigned to either the screen condition ($n = 25$, $M_{\text{age}} = 4.81$ years, nine girls, 16 boys) or window condition ($n = 26$, $M_{\text{age}} = 4.63$ years, 10 girls, 16 boys) condition semi-randomly, with the constraint that mean age was similar across groups.

2.2 | Setting and materials

Figures 1 and 2 show the experimental room layout. Curtains hung from the ceiling to the floor, separating the sandbox search area from the viewing area. The search area measured approximately 2.5 m square, with an opening in the curtain near the door where children entered to search. When the child entered the experimental room, the large sandbox was visible to their right and the viewing position was directly in front of them, with an area for demonstration of the task to the left. During hiding events, an experimenter sat centered along the long edge of the sandbox facing the child. A short table to the experimenter's right, not visible to the child, held a marker board for recording session information, a laptop that controlled the overhead video recording, and toys for hiding.

Children viewed hiding events from outside the curtains, either directly through a window in the middle of the curtain or via closed-circuit video feed on a computer monitor aligned within the window. We used a live video rather than a prerecorded video demonstration for two reasons. First, a live video feed ensured that the degree of subtle variations in demonstrations (e.g., how long it took the experimenter to hide the toy on each trial) was similar in the two conditions. Sec-

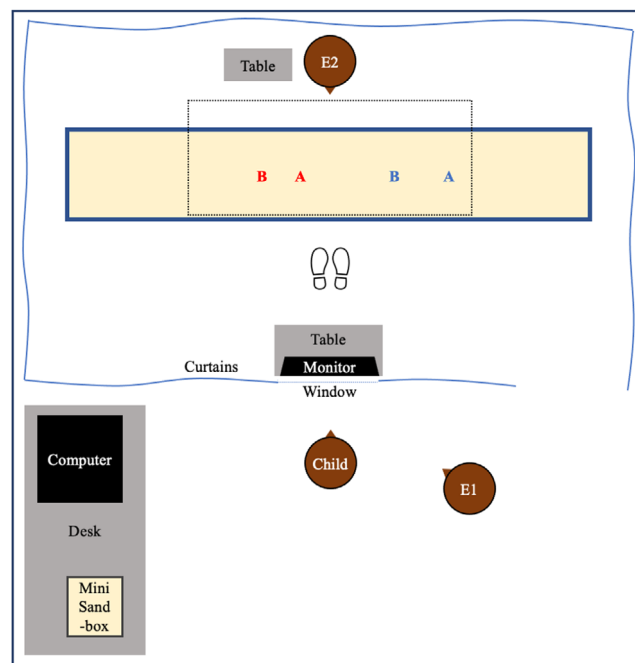


FIGURE 1 Schematic of the room layout viewed from above in the screen condition (not to scale). The layout was the same in the window condition except the monitor was removed, allowing the child to see E2 and the sandbox through the window in the curtain. E1 stood to the right of the child during hiding events. The dotted line indicates the approximate view of the sandbox as recorded by the overhead camera for subsequent coding. E2 sat on the far side of the sandbox, hiding toys in blocks of six A trials followed by six B trials. B was always 15.24 cm (6 in.) from the midline of the sandbox. A was always to the right of B (from the child's perspective). A was 7.62 cm (3 in.) from B for blocks on the left side of the sandbox and 15.24 cm (6 in.) from B for blocks on the right side of the sandbox.

ond, a live video feed helped to establish the real-time symbolic relation between the image on the screen and the experimenter and sandbox on the other side of the curtain (described later).

While viewing hiding events, children either stood on an outline of two footprints on the floor, stood on a short stool, or sat in a highchair, ensuring a similar viewing angle across children with different heights. Within the curtained search area, the sandbox was placed approximately 75 cm in front of the far wall. This placement allowed the full length of the sandbox to be visible through the window or on the video screen.

The testing sandbox was constructed from 2.5-cm thick plywood and measured 152 cm long by 41 cm wide by 51 cm high, with a false bottom 15 cm below the top (Schutte et al., 2011). The sandbox was painted dark blue and contained play sand approximately 10 cm deep. White paint, not visible to the participant, marked 15-cm intervals along the side of the sandbox facing the experimenter to facilitate finding the hiding locations. A metal bar mounted with wooden pegs at 2.5-cm intervals ran the length of the sandbox beneath the sand. Small fish toys, measuring approximately 1.25 cm wide by 5.5 cm long by 5 cm tall (see Figure 3a), were hidden by placing them into a paperclip attached to each wooden peg.



FIGURE 2 Room setup during the screen condition. A computer monitor was placed directly behind the curtain window to align the screen with the opening. The monitor displayed a live closed-circuit video feed from a camera mounted on the back of the monitor, showing hiding events as they occurred in real time. In the window condition, the computer monitor was removed so children could view hiding events directly through the curtain window.

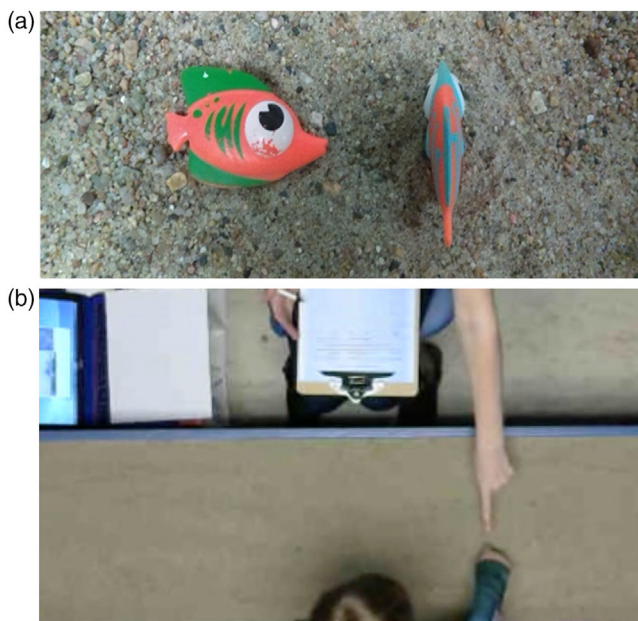


FIGURE 3 (a) Fish toys used in the search task. (b) View of the sandbox from the overhead camera used for subsequent coding.

A digital video camera was mounted on the ceiling directly over the testing sandbox to record each session for subsequent coding (described later). Figure 3(b) shows a screenshot from the overhead camera.

A miniature sandbox was used for training (described later), which took place in the back of the viewing area, facing away from the curtain window or monitor. The miniature sandbox was created from a rectangular foil pan filled with play sand. The foil pan measured 36 cm long by 28 cm wide by 8 cm high.

2.3 | Procedure

Two experimenters (E1, E2) conducted each session. E1 stayed with the child in the viewing area, described the hiding game, and presented training trials with the miniature sandbox. Meanwhile, E2 sat behind the large sandbox on the other side of the curtain and was visible through the window or on the video screen. To introduce the task, E1 said, “We’re going to play a sandbox game! Do you see my friend over there? What you’re going to do is watch my friend *through the window* [on the screen], and she’s going to hide the fish. But first, we’re going to practice our sandbox game in the little sandbox. Do you see the little sandbox? I have a little fish too!” Using the small training sandbox, E1 demonstrated the task while sitting on the floor with the child. On each of two training trials, E1 held up the fish and asked, “Do you see the fish?” E1 buried the fish in the sand while saying, “I’m going to cover this fish up.” After burying the fish, E1 said, “Now can you cover your eyes and count to five with me?” After counting together, E1 held up an index finger and asked, “Can you use one pointer finger and show me where you think the fish is?” Once the child pointed, E1 pointed as well, saying, “OK, is it right there? Let’s see where the fish is!” E1 helped the child dig for the fish, exclaiming, “There it is! Nice job! You found the fish!”

Following training, E1 asked, “Do you think you can do this in the big sandbox?” E1 led the child to the window (or video screen) in the viewing area and pointed to the footprint outlines, saying, “Can you stand on the feet right there?” E1 introduced the next part of the task by asking, “Do you see my friend *through the window* [on the screen] right there?” At this point, E2 waved and said, “Hi!” E1 then explained, “She’s going to hide a fish in the sandbox, just like we did over there, but you’re going to watch her *through the window* [on the screen].” Before starting each trial, E2 asked the child, “Do you see the fish?” If the child did not answer the question spontaneously, E2 prompted the child to answer out loud. E2 did not hide the fish until the child responded. This step was intended to establish contingency with E2 and correspondence between what was observed on screen and what was happening on the other side of the curtain. Once the child responded, E2 said, “Good. I’m going to hide him right here,” and buried the fish halfway in the sand so the top of the fish was still visible. E2 asked again, “Do you see the fish?” After the child responded, E2 said, “Now we’ll cover him up, and then we’ll close our eyes and count to five.” E2 covered the fish completely in sand and counted with the child with their eyes covered. Meanwhile, E1 watched



the child to ensure they saw the fish being hidden through the window (or video screen) and closed their eyes during the 5-s delay. After counting, E2 said, "Now can you come point to where you think the fish is?" E1 led the child to the large sandbox, and E2 said, "Can you stand on those feet?" while pointing to the footprint outlines on the floor. Once the child was in position, E2 held up an index finger and said, "Can you point with your pointer finger to where you think the fish is?" After the child pointed, E2 pointed to the same location to aid coding (see Figure 3b) and said, "Right here? OK, let's find the fish!" E2 helped the child dig for the toy, cheering "There is!" when they found the toy. Then E1 took the child back to the viewing area for the next trial.

2.4 | Search task design

The toy was hidden in one of four locations on each trial. Two locations were defined as "A" locations and two as "B" locations for an A-not-B manipulation (i.e., six A trials followed by six B trials, with the B trials analyzed for perseverative bias). Figure 1 shows the A and B locations on each side of the sandbox. The locations were chosen as likely to show perseverative bias in this age range (Schutte et al., 2003).

Each child completed one block (six A trials followed by six B trials) on the left side of the sandbox and another block on the right side of the sandbox, for a total of 24 trials. The two blocks were separated by a 10-min break during which children completed an unrelated task outside of the sandbox area. Block order (right side first vs. left side first) was counterbalanced across children and condition.

2.5 | Coding

Children's search locations were coded from overhead video recordings by one of four trained research assistants. Six of the 51 children in the final sample had codable data from one block only. Missing blocks were due to technical failure (2), experimenter error (1), or child non-compliance (3). Additionally, 27 of the remaining 1152 trials (2.3%) were excluded across 15 participants due to technical failure (three trials from one participant), experimenter error (nine trials from eight participants), noncompliance (two trials from one participant), or the child searching off-camera (13 trials from seven participants).

Coders were naïve to the hypotheses being tested, and the condition assigned to the participant was not identifiable from video recordings. If a coder identified trials or sessions as difficult to code, the second author served as the coder without knowledge of the condition assigned. A standard letter-size sheet of paper held in E2's lap (see Figure 3a) was used to adjust the scale of the sandbox as it appeared in the overhead video. That is, at the beginning of each video, coders measured the size the sheet of paper appeared on the computer screen, and this measurement was used in a formula to scale between the video image and actual size (i.e., error size on screen * paper actual size/paper size on screen). Transparent plastic sheets were taped over the computer monitor on which coders viewed the videos. On each trial, they used a dry-erase marker to

mark the hiding and search locations, then used a fabric tape measure to record the difference to the nearest 0.1 cm. Search errors were coded as positive to the right of the hiding location and negative to the left. Coders recorded search errors for each trial on a coding sheet, then entered these and the paper measurement into a spreadsheet.

Search locations were coded as the point in the sand where the child's index finger made first contact, unless the view was obscured by the child's head as they leaned over the sandbox, the child searched off-camera, or the first contact was clearly unintentional (e.g., in the beginning or middle of the pointing motion). For trials when the child's index finger first contact was not used, the location where E2 pointed to the sand after the child's point was used if possible. If E2's point was also outside of the camera view, which subtended approximately 60% of the sandbox with midline in the center (Figure 1), the trial was excluded as off-camera.

Videos for eight participants (15%) were coded by two independent observers to determine inter-rater reliability. Absolute agreement was high for children's search error on each trial, with an intra-class correlation of 0.94 (95% CI = 0.91–0.96; $F[92, 92.1] = 30.40, p < 0.001$).

2.6 | Analytical approach

Our primary research questions involved the impact of condition (window vs. screen) on children's search error in the absence of proactive interference (A trials) and the presence of proactive interference (B trials). A secondary research question was the extent to which condition effects and error distributions changed across trials within a block. Initial data exploration included plotting distributions of search errors and testing for normalcy. We used Kolmogorov–Smirnov (K–S) tests to assess the degree of overlap between the distributions of search errors in the window versus screen conditions and Levene's tests to compare the variability in these distributions. Finally, to test the extent of proactive interference on B trials (i.e., error toward the A location), we fit a multilevel linear model with trials nested within children. Fixed effects included condition (window vs. screen) and trial number within each block (1 through 6). Participant was a random effect. Child age (mean-centered) was a covariate. Models were estimated using the lmer function from the package lme4 (Version 1.1-18-1) (Bates et al., 2015) in the R software environment (Version 3.5.1) (R Core Team, 2018). The data and analysis scripts that support the findings of this study are available at <https://nyu.databrary.org/volume/1406>.

3 | RESULTS

Figure 4 plots individual search errors for each trial (up to 24 per child). The overall distribution of search errors was non-normally distributed, as evidenced by a significant Shapiro–Wilk test ($W = 0.84, p < 0.001$). Specifically, the distribution was kurtotic (kurtosis = 7.16), with most searches falling near the actual hiding location (errors near 0 cm) and relatively few searches falling far from the hiding location. This was

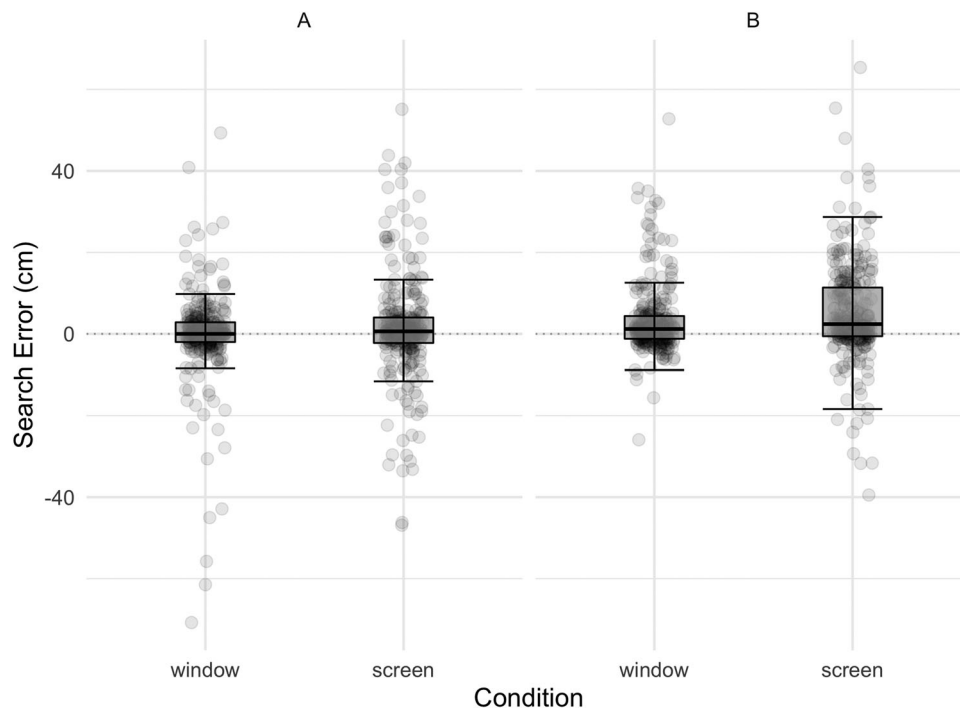


FIGURE 4 Distributions of children's search errors (cm) as a function of trial type (A vs. B) and condition (window vs. screen).

true regardless of trial type (A, B), trial order (first three, last three), and condition (window, screen) (kurtosis range = 3.04–16.20). Nonetheless, the same qualitative pattern of results was observed when using the entire data set versus a trimmed subset of data that excluded outliers, defined as $1.5 \times \text{IQR}$ (inter-quartile range) below the 25th percentile or above the 75th percentile. Thus, findings presented here are based on the entire data set. See Supplementary Materials for the subgroup statistics (Table S1) and robustness check (Table S2) described here.

We used K–S tests to assess the degree of overlap between the distributions of search errors in the window versus screen conditions (i.e., testing the alternative hypothesis that the window and screen data were drawn from different underlying distributions). Separate K–S tests were run for A and B trials. The first K–S test indicated that the window and screen errors were likely drawn from the same underlying distribution on A trials ($D = 0.08$, $p = 0.282$). However, a significant K–S test for B trials suggests that search errors in the two conditions were likely drawn from different underlying distributions ($D = 0.17$, $p < 0.001$). We used Levene's tests to compare the variability in search errors in each condition. Both tests were significant (A trials: $F[1, 562] = 5.58$, $p = 0.019$, B trials: $F[1, 559] = 19.60$, $p < 0.001$). Together, the results indicate that A-trial searches were more variable in the screen condition ($SD = 12.64$ cm) than the window condition ($SD = 11.08$ cm) but were drawn from the same underlying distribution. Conversely, B-trial searches were not only more variable in the screen condition ($SD = 12.30$ cm vs. 8.62 cm) but also drawn from a different underlying distribution.

Of particular interest in the current study was perseverative bias, or the extent to which proactive interference from the A trials influenced

children's performance on B trials. Within each block, the A location was always to the right of the B location. Therefore, B-trial errors that were greater than zero were errors in the direction of the A location. Overall, 68% of searches on B trials produced positive errors (i.e., in the direction of the A location), with a median error of 1.56 cm. A sign test indicated that B-trial errors were more likely to be positive than negative ($s = 355$, $p < 0.001$). Positive errors accounted for 64% of B-trial searches in the window condition and 72% of B-trial searches in the screen condition.

We used a linear multilevel model to examine perseverative bias on B trials as a function of condition, trial, and age. See Table 1 for a full report of fixed and random effects. The intercept was positive and significant, indicating a significant degree of perseverative bias (positive error in the direction of A) ($B = 9.02$, $SE = 1.61$, $p < 0.001$). There was a main effect of condition such that children exhibited greater perseverative bias in the screen condition than the window condition ($B = -1.14$, $SE = 0.33$, $p = 0.001$). There was also a significant effect of trial number such that children exhibited more perseverative bias earlier in each block than on later trials ($B = -4.78$, $SE = 2.27$, $p = 0.037$). The condition-by-trial interaction was not significant ($B = 0.84$, $SE = 0.47$, $p = 0.073$). Age was not a significant covariate ($B = -0.31$, $SE = 1.65$, $p = 0.852$). In sum, the model results provide evidence of perseverative bias, particularly in the screen condition and particularly at the start of each B block.

Figure 5 shows distributions of search errors for each B trial and condition. A visual inspection suggests unimodal distributions shifted slightly toward A, at least on early B trials, rather than bimodal distributions with some searches at location A and others at location B.

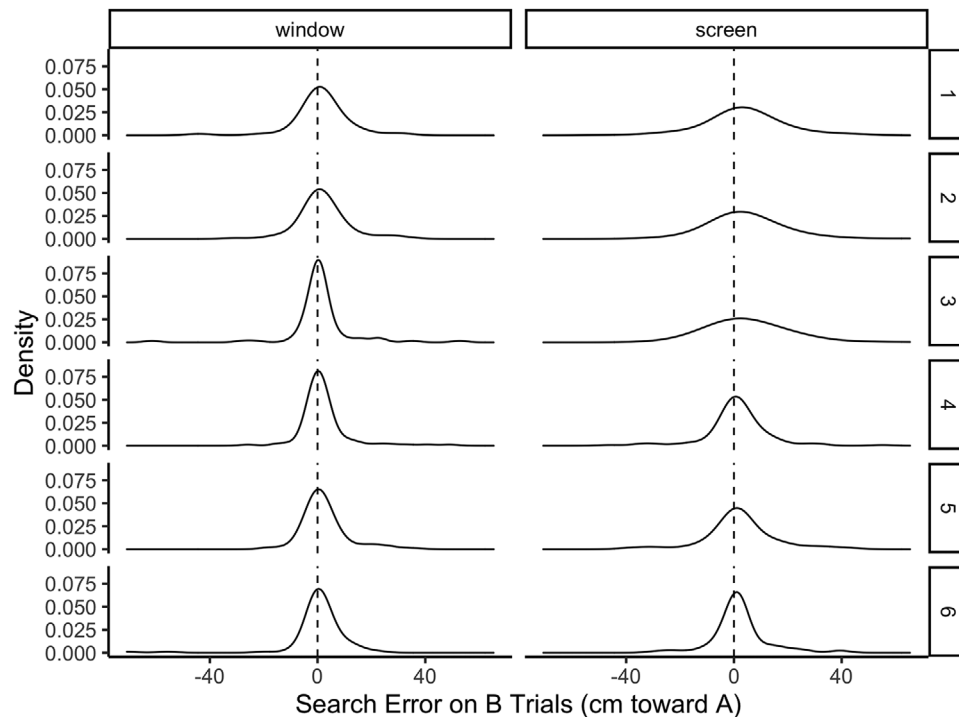


FIGURE 5 Kernel density estimates for distributions of search errors on B trials as a function of trial number and condition.

TABLE 1 Fixed and random effects for linear multilevel models predicting B-trial search errors as a function of age, trial number, and condition.

Predictors	Search error (cm)		
	Estimates	CI	p
(Intercept)	9.02	5.87 – 12.17	<0.001
Age (mean centered)	–0.31	–3.54 – 2.92	0.851
Trial number	–1.14	–1.79 – 0.50	0.001
Condition (window)	–4.78	–9.23 – 0.33	0.035
Trial number * Condition (window)	0.84	–0.08 – 1.76	0.073
Random effects			
σ^2	88.40		
τ_{00} Part.ID	24.07		
ICC	0.21		
$N_{\text{Part.ID}}$	51		
Observations	561		
Marginal R^2 / Conditional R^2	0.025 / 0.234		

4 | DISCUSSION

Most studies suggest the video deficit declines by 36 months of age, but there is some evidence of a video deficit among older preschoolers when using more complex learning tasks (Barr, 2010; Dickerson et al., 2013; Flynn & Whiten, 2008; Kirkorian, 2018; Reiß et al., 2019). The strongest evidence of a video deficit has been found in spatial

recall tasks (Strouse & Samson, 2021). Several not-mutually-exclusive accounts for the video deficit emphasize memory constraints, such as mismatched retrieval cues when transferring between different contexts (e.g., 2D to 3D) and competition between multiple memory representations when learning across multiple trials. Yet, while such accounts of the video deficit imply that memory representations are graded in nature rather than simply present or absent, the spatial recall tasks used to study the video deficit operationalize children's memory as binary (success or failure on each trial). Such studies are likely to mask gradations in children's performance.

Here, we use a continuous measure of spatial recall, allowing for behavioral evidence of gradations in memory representations. We posit that other spatial recall tasks mask a video deficit among older children, underestimating the age at which the video deficit disappears (Barr, 2010; Kirkorian, 2018). Specifically, we argue that weak representations formed from video may be sufficiently strong to support accurate searches in the spatial recall tasks typically used to study the video deficit (i.e., those with discrete, easily nameable hiding locations) but not more difficult spatial recall tasks that require stronger representations (Munakata, 2001). Using a more sensitive spatial recall task with a continuous search space, we tested whether 4- to 5-year-old children would show differences in memory strength when watching hiding events through a window versus video screen. The design of the task (a modified A-not-B paradigm) and the continuous measure of search performance allowed us to assess gradations in children's search performance as evidenced by the variability of search errors and degree of perseverative bias across trials.

The distributions of search errors were kurtotic, with most searches occurring on or near the location of the hidden toy. This was true in



both conditions. Nonetheless, we observed greater variability in children's search performance in the screen condition than in the window condition, even on A trials. Greater variability has also been observed among younger (vs. older) children in past research using an in-person version of another spatial recall task with a continuous search space, and computational modeling supported the interpretation that higher variability among younger children was due to relatively weak memory representations compared to those of older children (Schutte & Spencer, 2009; Simmering & Schutte, 2016). This interpretation supports our hypothesis that the screen condition resulted in weaker representation than the window condition in our study. Moreover, it suggests the performance of children in the screen condition may emulate performance of younger children in the window condition due to more variability in memory representations for location. This possibility could be tested experimentally in future behavioral and computational research.

On B trials, the observed distribution of search errors was not only more variable in the screen condition but was likely to be drawn from a different underlying distribution. This appears to be due to greater perseverative bias in the screen condition than the window condition. B-trial errors were more likely to occur toward than away from A, particularly in the screen condition. Moreover, a visual inspection of distributions on B trials highlights the utility of this task design for capturing continuous gradation in representations: unimodal distributions centered near (but not exactly at) B indicate that children in both conditions were representing the B location in memory, but there was more bias toward the A location in the screen condition than the window condition. By contrast, a bimodal distribution with separate modes at the A and B location would indicate that some children were forgetting the B location and searching back at A, analogous to the perseverative errors seen in spatial recall tasks with discrete hiding locations. Together, the findings suggest that perseverative bias is due to a memory representation that is built up across trials, integrating children's on-screen and in-person experience from previous trials, rather than a failure to choose a correct representation over a separate, outdated representation.

The increase in perseverative bias following screen demonstrations may seem counterintuitive insofar as children in the screen condition appear to have formed a weaker memory representation of the A location (i.e., more variable search errors). One might expect a relatively stronger representation in the window condition to produce more proactive interference than the relatively weaker representation in the screen condition. However, the repeated hiding events at the A location, along with searches distributed around the A location, seem to have built up a strong enough representation to "attract" search on B trials in both conditions, and this attraction was stronger than the representation of the B location in the screen condition. That is, the cumulative representation of the A location across trials (through both hiding and search events) could be stronger in the window condition than the screen condition, but because the representation of the B location was also stronger in the window condition, there was less evidence of proactive interference in the window condition.

The current study is limited by a small, homogenous convenience sample, necessitating replication. Moreover, the age range was relatively narrow, so future studies should test both older and younger children to establish the time course of the video deficit in this task. The study is also limited in its ability to pinpoint the precise mechanism(s) underlying the video deficit. Our behavioral findings are consistent with several existing accounts of the video deficit, most of which emphasize the role of memory constraints and imply memory representations are graded in nature. However, our findings cannot isolate specific memory constraints that produced a video deficit here. Among other things, we do not know whether the observed video deficit was due to general memory constraints inherent to any transfer task (e.g., mismatch between perceptual cues in the 2D encoding vs. 3D retrieval situation) or due to the nature of video itself (e.g., slower processing of perceptually impoverished 2D images compared to perceptually rich 3D experience). Future research could use our spatial recall task to probe questions about underlying mechanisms. For example, one could use this task to test cross-dimension transfer from video to real-life stimuli and vice-versa, analogous to the approach used by Zack and colleagues (2009) to test toddlers' cross-dimension transfer for imitation. Such a study could reveal whether our behavioral results reflect a specific video deficit affecting 2D but not 3D learning versus a general transfer deficit that appears in both contexts.

Despite the study's limitations, our findings demonstrate a video deficit for spatial recall that persists long after the third birthday. This is the first study demonstrating a protracted video deficit for spatial recall using a continuous measure that captures gradations in accuracy and perseverative bias on each trial. Our findings provide direct evidence that perseverative bias is graded, and memory representations are built up and strengthened with repetition. For these reasons, our findings help to explain why children may demonstrate a video deficit in some tasks or contexts but not others. Specifically, while a weak mental representation from video may be sufficient in a relatively simple task (e.g., the discrete locations used in prior spatial recall tasks), a similarly weak representation may not be enough in a more difficult task (e.g., our sandbox task). By extension, the video deficit should be protracted for more difficult tasks measuring the same knowledge but that require stronger memory representations. This possibility could be tested directly in future research using within-subject manipulations of task constraints (e.g., memory load, visual landmarks).

Given that transferring information from one context to another is fundamental to learning across the lifespan (Barnett & Ceci, 2002), the implications for early learning may be profound. Findings from this and other studies raise questions about the developmental time course for the video deficit, suggesting this effect may be protracted for more difficult tasks or when there are opportunities for proactive interference (Barr, 2010; Kirkorian, 2018; Troseth, 2010). If so, the upper bound of the video deficit has yet to be established, assuming such a bound exists. Paradigms that are sensitive to graded differences in representational strength may reveal more subtle video deficit effects than previously observed, with far-reaching, domain-general implications.

ACKNOWLEDGMENTS

Data collection and coding were conducted at the University of Wisconsin-Madison. Vanessa Simmering contributed to the manuscript while working for ACT, Inc., and is currently employed by the University of Kansas and Doctrina Consulting, LLC. Preliminary findings were presented alongside those of other studies at the 2018 Cognitive Science Society annual meeting and described in a 2-page summary in the published meeting proceedings. Findings were also presented at the 2022 annual meeting of the International Communication Association. We thank Chelsea Andrews and Hilary Miller for help implementing the design and supervising research assistants; the research assistants who collected data (Erin Daleccio, Sarah Habermehl, Kirsten Ippach, Dorie Ross, Samantha Savayong, Benjamin Smith, Emily Terlap, Emily Ticho) and coded videos (Danielle Erb, Riley Gramza, Kirsten Ippach, Emily Terlap); and the families who participated.

CONFLICT OF INTEREST STATEMENT

The authors report no potential conflicts of interest.

DATA AVAILABILITY STATEMENT

The data and analysis scripts that support the findings of this study are available at <https://nyu.databrary.org/volume/1406>.

REFERENCES

- Anderson, D. R., & Pempek, T. (2005). Television and very young children. *American Behavioral Scientist*, 48(5), 505–522. <https://doi.org/10.1177/0002764204271506>
- Barnett, S. M., & Ceci, S. J. (2002). When and where do we apply what we learn?: A taxonomy for far transfer. *Psychological Bulletin*, 128(4), 612–637. <https://doi.org/10.1037/0033-2909.128.4.612>
- Barr, R. (2010). Transfer of learning between 2D and 3D sources during infancy: Informing theory and practice. *Developmental Review*, 30(2), 128–154. <https://doi.org/10.1016/j.dr.2010.03.001>
- Barr, R. (2013). Memory constraints on infant learning from picture books, television, and touchscreens. *Child Development Perspectives*, 7(4), 205–210. <https://doi.org/10.1111/cdep.12041>
- Barr, R., Muentener, P., Garcia, A., Fujimoto, M., & Chávez, V. (2007). The effect of repetition on imitation from television during infancy. *Developmental Psychobiology*, 49(2), 196–207. <https://doi.org/10.1002/dev.20208>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Carver, L. J., Meltzoff, A. N., & Dawson, G. (2006). Event-related potential (ERP) indices of infants' recognition of familiar and unfamiliar objects in two and three dimensions. *Developmental Science*, 9(1), 51–62. <https://doi.org/10.1111/j.1467-7687.2005.00463.x>
- Dickerson, K., Gerhardstein, P., Zack, E., & Barr, R. (2013). Age-related changes in learning across early childhood: A new imitation task. *Developmental Psychobiology*, 55, 719–732. <https://doi.org/10.1002/dev.21068>
- Flynn, E., & Whiten, A. (2008). Imitation of hierarchical structure versus component details of complex actions by 3- and 5-year-olds. *Journal of Experimental Child Psychology*, 101(4), 228–240. <https://doi.org/10.1016/j.jecp.2008.05.009>
- Hipp, D., Gerhardstein, P., Zimmermann, L., Moser, A., Taylor, G., & Barr, R. (2017). The dimensional divide: Learning from TV and touchscreens during early childhood. In *Media exposure during infancy and early childhood* (pp. 33–54). Springer. https://doi.org/10.1007/978-3-319-45102-2_3
- Kirkorian, H. L. (2018). When and how do interactive digital media help children connect what they see on and off the screen? *Child Development Perspectives*, 12(3), 210–214. <https://doi.org/10.1111/cdep.12290>
- Kirkorian, H. L., Lavigne, H. J., Hanson, K. G., Troseth, G. L., Demers, L. B., & Anderson, D. R. (2016). Video deficit in toddlers' object retrieval: What eye movements reveal about online cognition. *Infancy*, 21(1), 37–64. <https://doi.org/10.1111/inf.12102>
- Krcmar, M. (2010). Can social meaningfulness and repeat exposure help infants and toddlers overcome the video deficit? *Media Psychology*, 13(1), 31–53. <https://doi.org/10.1080/15213260903562917>
- Krcmar, M., Grela, B., & Lin, K. (2007). Can toddlers learn vocabulary from television? An experimental approach. *Media Psychology*, 10, 41–63. <https://doi.org/10.1080/15213260701300931>
- Kuhl, P. K., Tsao, F.-M., & Liu, H.-M. (2003). Foreign-language experience in infancy: Effects of short-term exposure and social interaction on phonetic learning. *Proceedings of the National Academy of Sciences*, 100(15), 9096–9101. <https://doi.org/10.1073/pnas.1532872100>
- McCall, R. B., Parke, R. D., Kavanaugh, R. D., Engstrom, R., Russell, J., & Wycoff, E. (1977). Imitation of live and televised models by children one to three years of age. *Monographs of the Society for Research in Child Development*, 42(5), 1–94. <https://doi.org/10.2307/1165913>
- Munakata, Y. (2001). Graded representations in behavioral dissociations. *Trends in Cognitive Sciences*, 5(7), 309–315. [https://doi.org/10.1016/S1364-6613\(00\)01682-X](https://doi.org/10.1016/S1364-6613(00)01682-X)
- Piaget, J. (1954). *The construction of reality in the child*. Basic Books.
- R Core Team. (2018). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Reiß, M., Krüger, M., & Krist, H. (2019). Theory of mind and the video deficit effect: Video presentation impairs children's encoding and understanding of false belief. *Media Psychology*, 22(1), 23–38. <https://doi.org/10.1080/15213269.2017.1412321>
- Roseberry, S., Hirsh-Pasek, K., Parish-Morris, J., & Golinkoff, R. M. (2009). Live action: Can young children learn verbs from video? *Child Development*, 80(5), 1360–1375. <https://doi.org/10.1111/j.1467-8624.2009.01338.x>
- Schmidt, M., Crawley-Davis, A., & Anderson, D. (2007). Two-year-olds' object retrieval based on television: Testing a perceptual account. *Media Psychology*, 9(2), 389–409. <https://doi.org/10.1080/15213260701291346>
- Schmitt, K., & Anderson, D. (2002). Television and reality: Toddlers' use of visual information from video to guide behavior. *Media Psychology*, 4(1), 51–76. https://doi.org/10.1207/S1532785XMEP0401_03
- Schutte, A. R., Simmering, V. R., & Ortmann, M. R. (2011). Keeping behavior in context: A dynamic systems account of a transition in spatial recall biases. *Spatial Cognition & Computation*, 11(4), 313–342. <https://doi.org/10.1080/13875868.2011.579212>
- Schutte, A. R., & Spencer, J. P. (2009). Tests of the dynamic field theory and the spatial precision hypothesis: Capturing a qualitative developmental transition in spatial working memory. *Journal of Experimental Psychology: Human Perception and Performance*, 35(6), 1698–1725. <https://doi.org/10.1037/a0015794>
- Schutte, A. R., Spencer, J. P., & Schöner, G. (2003). Testing the dynamic field theory: Working memory for locations becomes more spatially precise over development. *Child Development*, 74(5), 1393–1417. <https://doi.org/10.1111/1467-8624.00614>
- Sharon, T., & DeLoache, J. S. (2003). The role of perseveration in children's symbolic understanding and skill. *Developmental Science*, 6(3), 289–296. <https://doi.org/10.1111/1467-7687.00285>
- Simmering, V. R., & Schutte, A. R. (2016). Developmental dynamics: The spatial precision hypothesis. In G. Schöner, J. P. Spencer, & DFT Research Group (Eds.), *Dynamic thinking: A primer on dynamic field theory* (1st ed., pp. 251–270). Oxford University Press.



- Spencer, J. P., Smith, L. B., & Thelen, E. (2001). Tests of a Dynamic Systems Account of the A-not-B Error: The Influence of Prior Experience on the Spatial Memory Abilities of Two-Year-Olds. *Child Development*, 72(5), 1327–1346. <https://doi.org/10.1111/1467-8624.00351>
- Strouse, G. A., & Samson, J. E. (2021). Learning from video: A meta-analysis of the video deficit in children ages 0 to 6 years. *Child Development*, 92(1), e20–e38. <https://doi.org/10.1111/cdev.13429>
- Strouse, G. A., & Troseth, G. L. (2008). "Don't try this at home": Toddlers' imitation of new skills from people on video. *Journal of Experimental Child Psychology*, 101(4), 262–280. <https://doi.org/10.1016/j.jecp.2008.05.010>
- Suddendorf, T. (2003). Early representational insight: Twenty-four-month-olds can use a photo to find an object in the world. *Child Development*, 74(3), 896–904. <https://doi.org/10.1111/1467-8624.00574>
- Suddendorf, T., Simcock, G., & Nielsen, M. (2007). Visual self-recognition in mirrors and live videos: Evidence for a developmental asynchrony. *Cognitive Development*, 22(2), 185–196. <https://doi.org/10.1016/j.cogdev.2006.09.003>
- Troseth, G. L. (2003). TV guide: Two-year-old children learn to use video as a source of information. *Developmental Psychology*, 39(1), 140–150. <https://doi.org/10.1037/0012-1649.39.1.140>
- Troseth, G. L. (2010). Is it life or is it Memorex? Video as a representation of reality. *Developmental Review*, 30(2), 155–175. <https://doi.org/10.1016/j.dr.2010.03.007>
- Troseth, G. L., Casey, A. M., Lawver, K. A., Walker, J. M. T., & Cole, D. A. (2007). Naturalistic experience and the early use of symbolic artifacts. *Journal of Cognition and Development*, 8(3), 309–331. <https://doi.org/10.1080/15248370701446772>
- Troseth, G. L., & DeLoache, J. S. (1998). The medium can obscure the message: Young children's understanding of video. *Child Development*, 69(4), 950–965. <https://doi.org/10.1111/j.1467-8624.1998.tb06153.x>
- Troseth, G. L., Flores, I., & Stuckelman, Z. D. (2019). When representation becomes reality: Interactive digital media and symbolic development. *Advances in Child Development and Behavior*, 56, 65–108. <https://doi.org/10.1016/bs.acdb.2018.12.001>
- Wellman, H. M., Cross, D., & Watson, J. (2001). Meta-analysis of theory-of-mind development: The truth about false belief. *Child Development*, 72(3), 655–684. <https://doi.org/10.1111/1467-8624.00304>
- Zack, E., Barr, R., Gerhardstein, P., Dickerson, K., & Meltzoff, A. N. (2009). Infant imitation from television using novel touch-screen technology. *The British Journal of Developmental Psychology*, 27, 13. <https://doi.org/10.1348/026151008X33470010.1348/026151008X334700>
- Zelazo, P. D., Sommerville, J. A., & Nichols, S. (1999). Age-related changes in children's use of external representations. *Developmental Psychology*, 35, 1059–1071. <https://doi.org/10.1037/0012-1649.35.4.1059>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Kirkorian, H., & Simmering, V. (2023). Searching in the Sand: Protracted Video Deficit in U.S. Preschoolers' Spatial Recall Using a Continuous Search Space. *Developmental Science*, 26, e13376. <https://doi.org/10.1111/desc.13376>